Exercise 7 – Evaluation III

Introduction to Machine Learning

Exercise 1: k-NN

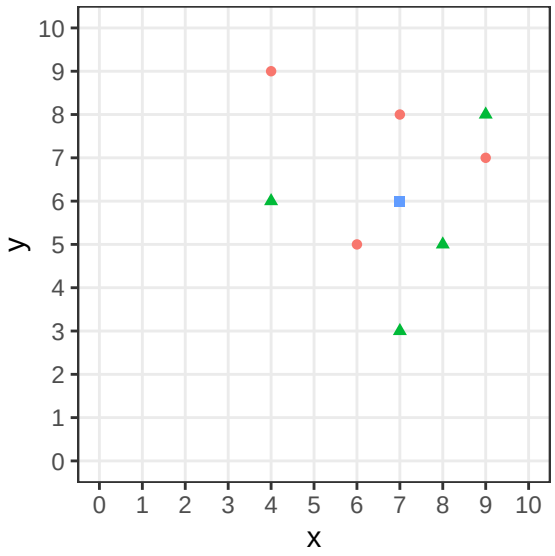
> Learning goals
>
> 1. Perform k-NN by visual means
> 2. Perform k-NN with pen and paper, possibly using weighted distances

Let the two-dimensional feature vectors in the following figure be instances of two different classes (triangles and circles).

a) Classify the point (7, 6) – represented by a square in the picture – with a k-NN classifier using $L1$ norm (Manhattan distance):

$$d_{\text{Manhattan}}(\mathbf{x}, \tilde{\mathbf{x}}) = \sum_{j=1}^{p} |x_j - \tilde{x}_j|.$$

As a decision rule, use the unweighted number of the individual classes in the k-neighborhood, i.e., assign the point to the class that represents most neighbors.



i. $k = 3$

ii. $k = 5$

iii. $k = 7$

b) Now consider the same constellation but assume a regression problem this time, where the circle-shaped points have a target value of 2 and the triangles have a value of 4.

Again, predict for the square point (7, 6), using both the *unweighted* and the *weighted* mean in the neighborhood (still with Manhattan distance).

Hint

We now consider both *unweighted* and *weighted* predictions. Recall that weights are computed based on the distance between the point of interest and its respective neighbors. With the Manhattan, or "city block" metric, the distance can be read from the plot by walking along the grid lines (shortest way). For example, in the 3-neighborhood, all points have a distance of 2 from our square, so all get weights $\frac{1}{2}$.

i. $k = 3$

ii. $k = 5$

iii. $k = 7$

Exercise 2: Classification metrics

Learning goals

1. Create confusion matrices and compute associated evaluation metrics
2. Compute ROC coordinates and AUC
3. Understand relationship between ROC curve & classification threshold
4. Assess probabilistic predictions using the Brier score

Consider a binary classification algorithm that yielded the following results:

ID	True class	Prediction
1	0	0.33
2	0	0.27
3	0	0.11
4	1	0.38
5	1	0.17
6	1	0.63
7	1	0.62
8	1	0.33
9	0	0.15
10	0	0.57

a) Create a confusion matrix assuming a threshold of 0.5. Point out which values correspond to true positives (`TP`), true negatives (`TN`), false positives (`FP`), and false negatives (`FN`).

b) Calculate: PPV, NPV, TPR, FPR, ACC, MCE and $F1$ measure.

c) Draw the ROC curve and interpret it.

d) Calculate the AUC.

e) How would the ROC curve change if you had chosen a different threshold in a)?

f) So far, we have evaluated the predicted *labels*. Now assess the predicted *probabilities* directly: compute the Brier score

$$\rho_{BS} = \frac{1}{n} \sum_{i=1}^{n} \left(\widehat{\pi}^{(i)} - y^{(i)} \right)^2 .$$

Does the result indicate a good probabilistic model? What advantage does this kind of evaluation have over label-based metrics?

Hint

To judge the magnitude of the Brier score, compare it to that of a naive baseline model.

Exercise 3: Choosing classification metrics

The whole exercise is only for lecture group B!

Learning goals

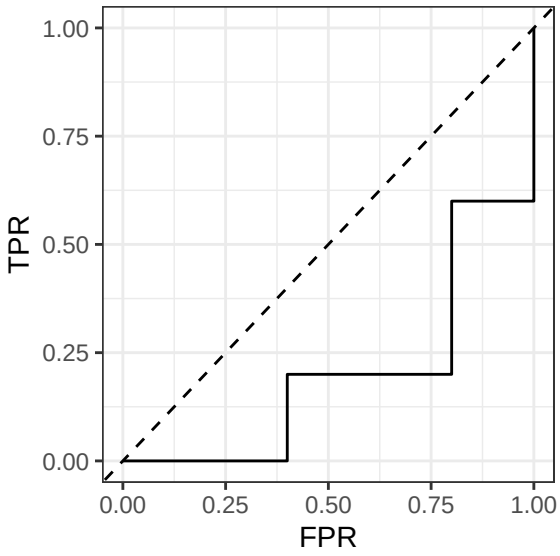
1. Understand why accuracy can be misleading for imbalanced data
2. Choose suitable evaluation metrics for different application scenarios
3. Sketch and interpret characteristic ROC curves

a) A hospital uses a classifier to detect a rare disease. Out of 1000 patients in the test data, only 20 actually have the disease. Consider a "lazy" classifier that simply predicts "healthy" for *every* patient. Compute the accuracy and the TPR of this classifier. Why is accuracy a misleading performance metric in this situation?

b) For each of the following scenarios, discuss which type of error – false positives or false negatives – is more critical, and which ROC metric you would therefore pay the most attention to:

   i. cancer screening ("positive" = cancer detected),
   ii. spam filter that moves mails flagged as spam ("positive") to the junk folder.

c) A colleague trained a scoring classifier for a similar task and evaluates it with the ROC curve shown below, obtaining an AUC of 0.2.



i. Your colleague is devastated. How would you assess the quality of this classifier?
ii. Suggest a way to improve the classifier – without any retraining.
iii. Sketch the ROC curve of the improved classifier. What is its AUC?

Exercise 4: Partial AUC

The whole exercise is only for lecture group A!

Learning goals

1. Understand why sometimes only a region of the ROC curve is practically relevant
2. Compute and interpret the (corrected) partial AUC

Reconsider the classifier from Exercise 2. Suppose it is used in a medical screening application where each positive prediction triggers an expensive follow-up examination. Since the clinic can only handle a limited number of these, operating the classifier at a false positive rate above 0.2 is not acceptable.

a) Explain why the full AUC from Exercise 2 d) may be a misleading criterion for choosing a classifier in this situation.

b) Compute the partial AUC over the region FPR ≤ 0.2 for the ROC curve from Exercise 2 c).

c) The raw pAUC is difficult to interpret because its range depends on the chosen region. Therefore, compute the corrected pAUC

$$\mathrm{pAUC}_{\mathrm{corrected}} = \frac{1}{2}\left(1 + \frac{\mathrm{pAUC} - \mathrm{pAUC}_{\mathrm{min}}}{\mathrm{pAUC}_{\mathrm{max}} - \mathrm{pAUC}_{\mathrm{min}}}\right),$$

where $\mathrm{pAUC}_{\mathrm{min}}$ and $\mathrm{pAUC}_{\mathrm{max}}$ denote the pAUC of a random and a perfect classifier over the same region, respectively. Interpret the result, also in comparison to the full AUC from Exercise 2 d).